MILITARY SPECIFICATION

BASE, TRAVERSING, JACK: 50-TON JACK

1. SCOPE

1.1 This specification covers a hand-operated, screw-type traversing base.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Box, Fiberboard.

Military

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

Federal

FED. TEST METHOD STD. NO. 101	- Preservation, Packaging, and Packing Materials: Test Procedures.

Military

MIL-STD-10	- Surface Roughness Waviness and Lay.
MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A47 - Malleable Iron Castings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N. Y. 10016.)

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, 1616 P Street NW, Washington, D. C. 20036.)

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3. REQUIREMENTS

3.1 Description. The traversing base shall conform to the dimensions shown on figure 1. The traversing base shall be capable of moving a 50-ton load through a horizontal distance of 15 to 18 inches. It shall consist of the following major components: a supporting base, a jack plate which slides upon the supporting base, a drive screw mounted in the supporting base and used to move the jack plate, ratchet mechanism to actuate the drive screw, and a wooden or steel operating lever.

3.2 Preproduction model. As soon as practicable after award of a contract, the supplier shall furnish a traversing base to prove, prior to starting production, that his production methods and choice of design detail will produce traversing bases that comply with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations from the preproduction model during production shall be subject to the approval of the contracting officer. Approval of the preproduction model by the activity concerned shall not relieve the supplier of his obligation to furnish traversing bases conforming to this specification.

3.3 Initial production traversing base. An initial production traversing base selected at random by the Government from the first five traversing bases being produced from production tooling shall be examined and tested as specified in 4.4 to determine conformance to the requirements of this specification. Acceptance of the initial production traversing base shall not establish the acceptability of the initial production run and the remainder of the traversing bases under contract in conflict with the acceptance levels specified in section 4. The initial production traversing base shall be fully representative of the traversing bases proposed to be furnished for the duration of the contract.

3.4 Material. Materials shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.5 Supporting base and jack plate. The supporting base shall be a malleable iron or low carbon steel casting or, a steel weldment. When the supporting base is a steel weldment it shall be stress-relieved before machining. The jack plate shall be of malleable iron castings conforming to ASTM A47, grade optional. The sliding surfaces of the base and load plates shall be machined to a surface roughness of not more than 63 in accordance with MIL-STD-10.

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3.6 Drive screw. The drive screw shall be fabricated from medium carbon steel. The screw threads shall be stub acme threads conforming to Handbook H28. Each end of the screw shall be machined to fit the ratchet mechanism.

3.7 Ratchet mechanism. The ratchet shall be of steel and the ratchet holder of malleable iron or steel. The pawl shall be of heat-treated steel and shall engage the teeth of the ratchet gear. With the traversing base supporting a 50-ton load, the ratchet mechanism when mounted on either end of the drive screw shall be capable of rotating the drive screw in both the clockwise and counterclockwise directions without damage or excessive wear of the pawl or of the ratchet teeth.

3.8 Operating lever. The operating levers shall be 24 inches long. The maximum diameter of the operating lever shall be 2 inches for wooden levers and 1 inch for steel levers. One end of the wooden lever shall be tapered to fit the socket of the ratchet. The wooden lever shall be fabricated from sound hickory, pecan, maple, oak, sycamore, or walnut. The wood shall be free of knots, wormholes, splits, wane, splinters, checks, or other injurious defects. The moisture content of the wood shall be between 10 and 14 percent, inclusive. The wood shall have straight grains; however, slight dips in the grain shall be acceptable. Operating the traversing base under full load shall not cause damage, failure or permanent deformation of the lever.

3.9 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with an adequate means of preventing loss of proper tightness and adjustment.

3.10 Screw-threads. Screw-threads shall conform to Handbook H28.

3.11 Bushings. Wherever bushings are required, the bushings shall be made of bronze, malleable iron, or of steel with bronze thrust washers.

3.12 Lubrication. All surfaces requiring periodic lubrication shall be provided with accessible means of lubrication. Each traversing base shall be lubricated.

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3.13 Identification marking. The traversing base shall be identified as shown on figure 1.

3.14 Treatment and painting. The traversing base shall be cleaned, treated and painted in accordance with MIL-T-704, type A.

3.15 Repair parts. Such repair parts as are specified shall be furnished (see 6.2).

3.16 Workmanship. All castings, forgings, stampings, or welded parts shall be free of harmful, extraneous materials. All iron castings shall be of uniform quality, free of blowholes, porosity, hard spots, shrinkage defects, cracks or other injurious defects. All components shall be free of burrs and sharp edges.

3.16.1 Interchangeability. All parts having the same manufacturer's part number shall be constructed to standards, tolerances, and clearances in order that such parts may be interchangeable, replaced, or adjusted without requiring modification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection of the traversing base (see 4.3).
(b) Initial production inspection of the traversing base (see 4.4).
(c) Quality conformance inspection of the traversing base (see 4.5).
(d) Inspection of preparation for delivery (see 4.7).

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4.3 Preproduction inspection.

4.3.1 Examination. The traversing base shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The traversing base shall be tested as specified in 4.6.2. Failure of the test shall be cause for rejection.

4.4 Initial production inspection.

4.4.1 Examination. The initial production traversing base shall be examined for the defects listed in 4.6.1.

4.4.2 Tests. Upon successful completion of the examination specified in 4.4.1, the initial production traversing base shall be subjected to the test specified in 4.6.2.

4.4.3 Inspection failure. Failure of the initial production traversing base to meet any requirement specified herein, during and as a result of the examination and tests specified in 4.4.1 and 4.4.2, respectively, shall be cause for refusal by the Government to accept production traversing bases until evidence has been provided by the supplier that corrective action has been taken to eliminate deficiencies.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105.

4.5.2 Examination. Samples selected in accordance with 4.5.1 shall be examined as specified in 4.6.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5.3 Test. Samples selected in accordance with 4.5.1 shall be tested as specified in 4.6.2. AQL shall be 2.5 percent defective.

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4.6 Inspection schedule.

4.6.1 Examination. The traversing bases shall be examined for the following defects:

Major

101. Dimensions not as specified.
102. Materials not as specified.
103. Components missing or not as specified.
104. Screw threads not as specified.
105. Castings, forgings, or stampings containing harmful extraneous material.
106. Iron castings containing blowholes, porosity, hard spots, shrinkage defects or cracks.
107. Lever not as specified or containing wanes, splits, checks, or other injurious defects.
108. Surface finish not as specified.
109. Components containing sharp edges or burrs.
110. Parts not interchangeable.

Minor

201. Traversing base not lubricated.
202. Identification marking missing, incorrect, or illegible.
203. Treatment and painting not as specified.

4.6.2 Test. Place a 50-ton load upon the jack plate of the traversing base. Mount the ratchet mechanism on one end of the drive screw and move the jack plate with respect to the base through the full travel once in each direction. Remove the ratchet mechanism and place it on the opposite end of the drive screw and again move the jack plate through the full travel once in each direction. Remove the load and examine the traversing base. Inability of the jack plate to move the full travel in each direction under load and damage to the ratchet mechanism, ratchet lever, or any component of the traversing base shall constitute failure of this test. Remove the ratchet mechanism and examine the ratchet and pawl. Damage or excessive wear of the pawl or of the ratchet teeth shall constitute failure of this test. Examine the lever. Failure of the lever, permanent deformation, cracked or split levers shall constitute failure of this test.

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4.7 Inspection of preparation for delivery.

4.7.1 Pilot-pack inspection.

4.7.1.1 Examination. Examine the pilot pack in accordance with 4.7.2.3. Presence of one or more defects shall be cause for rejection.

4.7.1.2 Test. The pilot pack for levels A and B shall be subjected to edgewise drop test, method 213, of STD- NO. 101. At the conclusion of the pilot pack test, the pack shall be opened and examined along with the contents. Rupture of the pack, loosening or breaking of holddowns, ties, stays, blocking, bracing, or any visible damage to the contents shall be cause for rejection.

4.7.2 Quality conformance inspection of pack.

4.7.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.2.2 Sampling. Samples for examination shall be selected in accordance with MIL-STD-105. AQL shall be 2.5 percent defective.

4.7.2.3 Examination. Samples selected in accordance with 4.7.2.2 shall be examined for the following defects.

110.	Materials not as specified.
111.	Preservation not as specified.
112.	Operating lever not secured to the traversing base.
113.	Ratchet mechanism not positioned to allow packing in a box of the smallest cube.
114.	Securing not adequate to prevent movement.
115.	Strapping not zinc coated for level A.
116.	Marking illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. The pilot pack is defined as a preproduction pack which will be identical to the pack which the supplier will subsequently produce in fulfillment of the contract. A pilot pack shall be made, examined, and subjected to the test to determine conformance to the preservation, packaging, packing, and marking requirements. Approval of the pilot pack shall not relieve the

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supplier of his obligation to preserve, package, pack, and mark the traversing bases in conformance to this specification. Any changes or deviations of production packs from the approved pilot pack shall be subject to the approval of the contracting officer.

5.2 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.2.1 Level A. Unpainted exterior metal surfaces of the traversing base, such as the drive screw and ratchet mechanism, requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative. The preservative shall conform to MIL-P-116. The operating lever shall be removed and secured to the traversing base. The ratchet mechanism shall be positioned in a manner to allow packing the traversing base in a box of the smallest cube.

5.2.1.1 Intermediate packaging. The preserved repair parts shall be intermediate packaged in a close-fitting box conforming to PPP-B-636, W5c.

5.2.2 Level C. The traversing base shall not require preservation. The ratchet mechanism shall be positioned in a manner to allow packing the traversing base in a box of the smallest cube. Repair parts shall be preserved and intermediate packaged as specified in 5.2.1 and 5.2.1.1 for level A.

5.3 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.3.1 Level A. Each traversing base and intermediate package shall be packed in a close-fitting box conforming to PPP-B-621, class 2, style optional or PPP-B-601, overseas type, style optional. The traversing base and intermediate package shall be secured in the box to prevent movement. The box shall be closed and strapped in accordance with the appendix to the applicable box specification. Strapping shall be zinc coated.

5.3.2 Level B. Each traversing base shall be packed as specified for level A except the box shall be domestic type or class as applicable and strapping need not be zinc coated.

5.3.3 Level C. Each traversing base shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates in a container complying with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other carrier rules applicable to the mode of transportation.

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5.4 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The traversing base is intended for use in erecting heavy fixed bridges. It is used to move heavily loaded jacks or the ends of a bridge through short distances in aligning the bridge with the abutment.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When repair parts are required (see 3.15).
(c) Level of preservation and packaging and level of packing required (see 5.2 and 5.3).

6.3 Supersession data. This specification supersedes in part MIL-B-12745, dated 22 May 1953 with Amendment 1, dated 10 February 1954.

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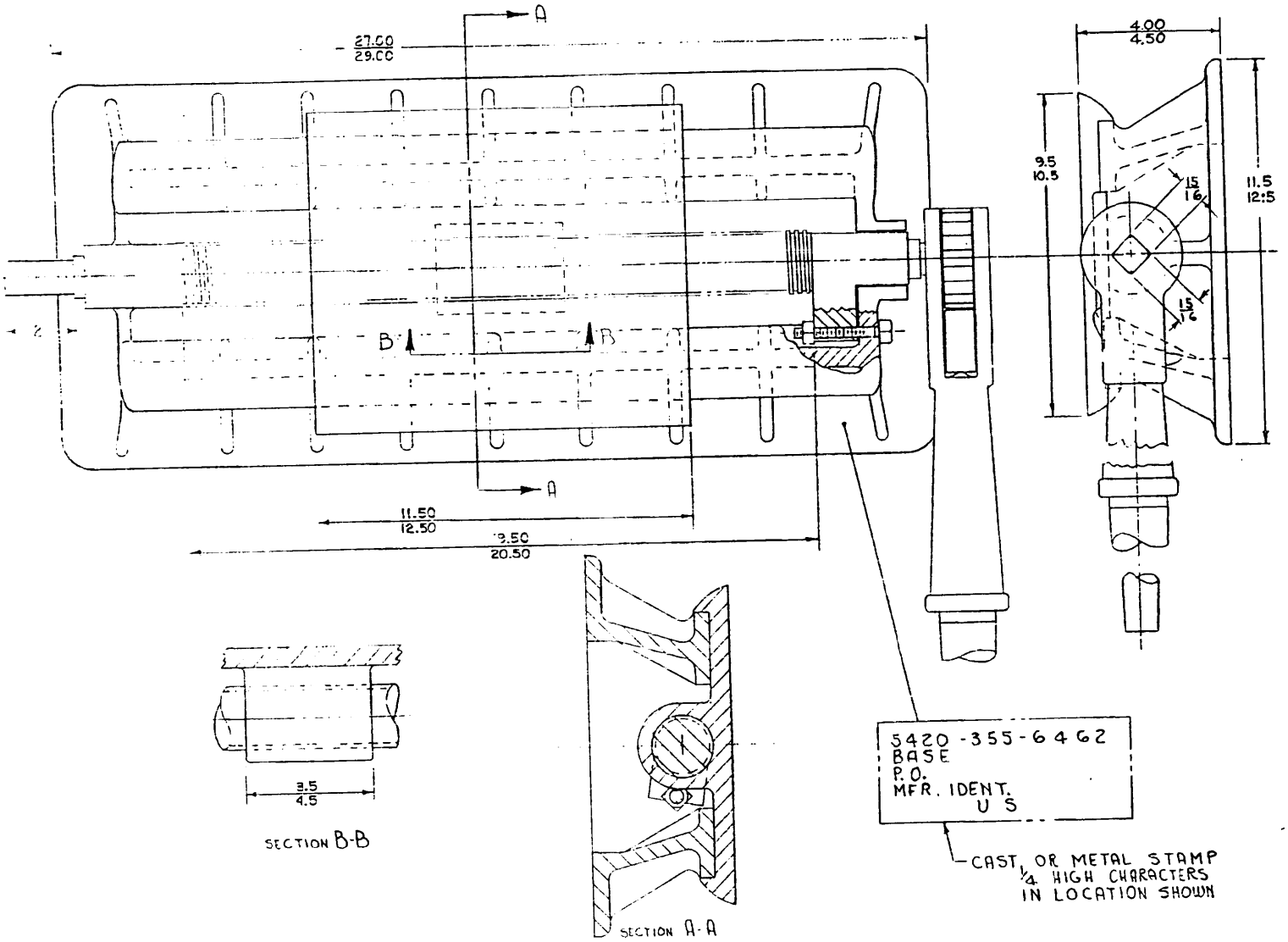
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NOTES:
1. ALL DIMENSIONS ARE IN INCHES

FIGURE 1. BASE, TRAVERSING JACK

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-B-52426(MO) - Base, Traversing, Jack: 50- Ton Jack

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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